MACHINE LEARNING

A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

Computer Engineering Department

Shankersinh Vaghela Bapu Institute of Technology, Gandhinagar

Gujarat Technological University, Ahmedabad

April 2025

Shankersinh Vaghela Bapu Institute of Technology

Gandhinagar-Mansa Highway, Vasan, Gandhinagar–382650

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **House Price Prediction** has been carried out by **Rajpurohit Pravinsingh V** under my guidance in partial fulfillment for the degree of Bachelor of Engineering in **Computer Engineering Department**, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2024-25.

Prof. Nirali Patel
Internal Guide

Prof. Samina Mansuri
Head of Department

Completion Certificate

Training Letter Head

Date:05/04/2025

TO WHOM IT MAY CONCERN

This is to clarify that Rajpurohit Pravinsingh V, a student of Shankersinh Vaghela Bapu Institute of Technology has successfully completed his Training Program in the field of Machine Learning from 19-01-2025 to 19-04-2025 (Total Number of Weeks:-12) under the guidance of Kishore.

During the period of his Training with us, he had been exposed to different processes and was found diligent, hardworking and inquisitive.

We wish him every success in his life and career.

For 1Stop.ai

Authorised Signature with Industry Stamp

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BANGALORE

Shankersinh Vaghela Bapu Institute of Technology

Gandhinagar-Mansa Highway, Vasan,Gandhinagar–382650

DECLARATION

We hereby declare that the Internship report submitted along with the Internship project entitled **House Price Prediction** submitted in partial fulfillment for the degree of Bachelor of engineering in Computer Engineering to Gujarat Technological University, Ahmedabad, is a bonafide record of original work carried out by me at 1stop.ai under the supervision of Seema Reshmi and that no part of this report has been directly copied from any students' report or taken from any other source, without providing due reference.

Rajpurohit Pravinsingh V
Name of Student

Sign of Student

ACKNOWLEDGEMENT

II wish to express my sincere gratitude to my External guide Seema Reshmi for continuously guiding me at the company and answering all my doubts with patience. I would also like to thank Prof. Samina Mansuri (H.O.D. of BE, BE(CE) / IT Department) for motivating me every time whenever I get confused, I would also like to thank my Internal Guide Prof. Nirali Patel for helping me through my internship by giving me the necessary suggestions and advices along with their valuable co-ordination in completing this Internship.

I also thank my parents, friends and all the members of the family for their precious support and encouragement which they had provided in completion of my work. In addition to that, I would also like to mention the company personals who gave me the permission to use and experience the valuable resources required for the Internship.

Thus, in conclusion to the above said, I once again thank the staff members of 1stop.ai for their valuable support in completion of the Internship.

ABSTRACT

Accurate prediction of house prices is a key challenge in the real estate industry, where numerous factors such as location, property size, age, number of rooms, proximity to amenities, and neighborhood quality influence the market value of a property. This project presents a comprehensive machine learning approach to house price prediction using structured data. The process begins with data collection from various reliable sources, followed by data cleaning, preprocessing, and exploratory data analysis to identify patterns and correlations. Feature engineering is performed to enhance model input quality, and multiple machine learning algorithms—such as Linear Regression, Random Forest, and Gradient Boosting—are applied and compared. The models are evaluated based on performance metrics like RMSE (Root Mean Square Error) and R^2 score to ensure optimal prediction accuracy. After fine-tuning, the best-performing model is selected, deployed, and integrated into a user-accessible interface for real-time predictions. The project not only provides valuable insights into the housing market trends but also demonstrates the potential of data-driven decision-making in property valuations

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1. INTRODUCTION

COMPANY OVERVIEW:

Company Name:1stop.ai

Contact No: 7619186427

Email Id: hrteam@movidu.in

Website: https://www.1stop.ai

External Guide Name: Seema Reshmi

Brief Introduction about Company:

1Stop.ai is a technology-driven company that specializes in artificial intelligence (AI) solutions. Their core mission is to make AI more accessible, practical, and scalable for businesses across a variety of industries. 1Stop.ai is a forward-thinking artificial intelligence company dedicated to making AI accessible and impactful for businesses of all sizes. The company provides a comprehensive AI-as-a-Service (AIaaS) platform that enables organizations to seamlessly integrate artificial intelligence into their operations without requiring deep technical expertise.

1Stop.ai is a leading Indian EdTech company focused on transforming traditional education through a practical, industry-aligned learning model. The company operates at the intersection of education and technology, aiming to bridge the skills gap between academia and the rapidly evolving demands of the tech industry. The platform is especially popular among engineering students and early-career professionals across India, offering flexible remote internships, certified training programs, and expert mentorship. With partnerships that include prestigious institutions like IIT Roorkee and IIT Bombay, the company brings real credibility and high-impact learning opportunities to its user base

PROJECT OVERVIEW:

- This project focuses on developing a robust predictive model for House price prediction, aiming to provide accurate forecasts of sales volumes for various real-estate products.

- Leveraging data analysis and machine learning techniques, I aim to address the challenges posed by the dynamic nature of the house market, including seasonality, market trends, and promotional activities.

- By analysing comprehensive datasets comprising historical sales records, product attributes, market dynamics, and external factors, I seek to uncover in sights into sales patterns and drivers.

- The project encompasses key stages such as data collection, exploratory data analysis (EDA), feature engineering, model building, evaluation, and deployment.

- Ultimately, the goal is to empower stakeholders in the real-estate industry with actionable insights and predictive capabilities to optimise inventory management, resource allocation, and decision-making processes.

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PROJECT GOAL:

- The goal of the House sales prediction project is to develop a predictive model that accurately forecasts house sales based on historical data and relevant features.

- The ultimate objective is to assist real-estate companies in optimizing their sales strategies, inventory management, and resource allocation, thereby enhancing efficiency and profitability in the industry.

- By leveraging advanced machine learning techniques, the project aims to provide actionable insights into sales trends, customer behaviour, and market dynamics, ultimately contributing to informed decision-making and improved business performance in the real-estate sector.

PROJECT OBJECTIVES:

- Develop a user-friendly interface tailored for customer who wants to buy house , facilitating streamlined inventory management, efficient resource allocation, and strategic planning.
- Generate various types of visualisations, including line charts, bar charts, and scatter plots, to enable in-depth analysis and decision-making. The report and statistical analysis provided will assist vendor.
- These features may include predictive analytics for sales forecasting, identifying market trends, and optimising pricing strategies.
- Implement advanced machine learning algorithms to enhance predictive analytics capabilities, allowing for accurate forecasting of future sales demand based on historical data and relevant features.
- Enable the calculation of sales metrics and optimal order quantities for future stock management.
- Deploy the predictive model in a production environment, integrating it with existing sales forecasting systems and workflow

2. IMPLEMENTATION STAGE

Project Flow:

Step1:Gather historical sales data from various sources, including internal database and external market research reports.

Step2: Clean and process data to handle missing value, outliers and inconsistencies.

Step3: Analyse the distribution of sales data overtime then identify correlations between sales and various factors such as seasonality etc.

Step4:Feature Engineering is applied to extract relevant features from the dataset and engineer new features of external factors of sales.

Step5: Evaluate different machine learning algorithms such as linear regression,regularization,rideg regression

Step6:Compare different models and select the final ridge regression model.

Step 7: Deploy Machine Learning Model using Python Libraries such as Streamlit.

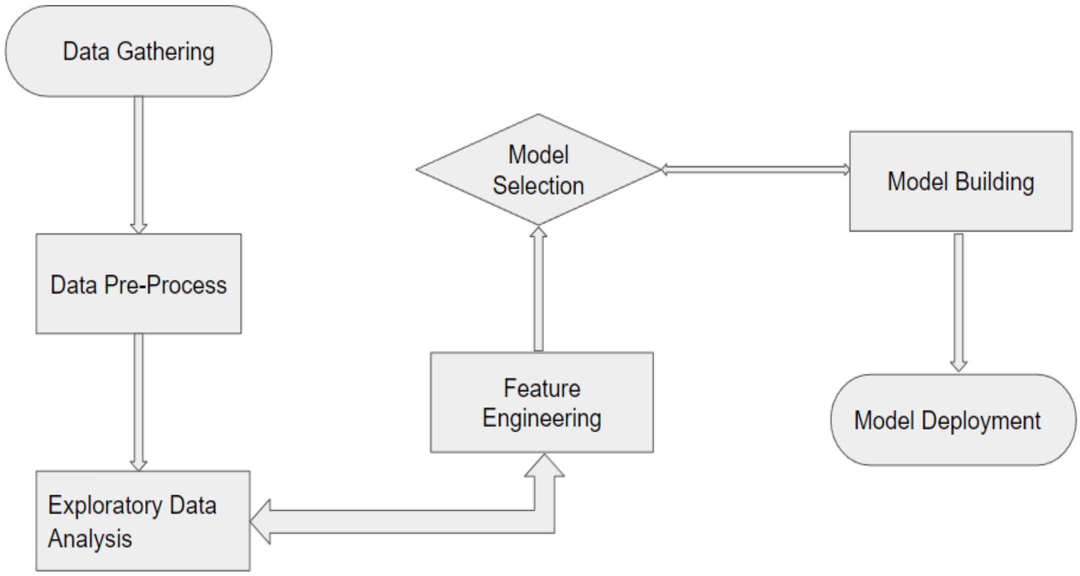


Figure2.1Process Flowchart

3. PROGRESS WORK

DATABASE:

DATACOLLECTION:

- Gather historical sales data from various sources and collect Additional data that may influence sales.
- Employ the Pandas library to manage the collected data effectively. Pandas facilitates the ingestion, manipulation, and analysis of structured data, ensuring seamless integration and processing.
- Pandas library is used to show gathered data which reads the contents of a CSV file into a DataFrame.

DATABASE EXPLANATION:

- **beds**
  - Number of bedrooms in the house. Typically correlates with house size and price.
- **baths**
  - Number of bathrooms in the house. Includes both full and partial baths.
- **size**
  - Size of the house in square feet. A major factor affecting price.
- **zip_code**
  - The ZIP code where the property is located. Useful for capturing location-based price variation.
- **price**
  - Target variable: The sale price of the house in rupees.

Column Name	Description	Type
Bedroom	Number of bedrooms	integer
Bathroom	Number of bathroom	float64
size	Area of house	float64
Zip_code	Postal code of city	Float64
price	Price of house	Float64

Table3.1DatabaseExplanation

DATA PRE-PROCESSING:

- Identify and address missing values in the dataset by either replacing them with appropriate values (such as the mean, median, or mode) or removing the rows or columns containing missing data. This ensures that missing values do not affect the analysis.
- Detects outliers in the dataset that may significantly deviate from the rest of the data and potentially skew the analysis. Outliers can be identified using statistical methods such asz-score or visualization techniques like boxplots .Once identified, outliers can be either removed from the dataset or adjusted appropriately to minimize their impact on the analysis.

```
data.isna().sum()

beds              0
baths             0
size              0
size_units        0
lot_size        347
lot_size_units  347
zip_code          0
price             0
dtype: int64
```

```
df.duplicated().any()

False
```

Figure 3.2 Duplicate Value Verification

- Check the dataset for duplicate values, which can distort the analysis and lead to biassed results. Duplicate values may arise due to data entry errors or system glitches. It's essential to identify and eliminate duplicate values from the dataset to ensure the accuracy and integrity of the analysis.

EXPLORATORY DATA ANALYTICS:

- Exploratory Data Analysis (EDA) serves as the foundational step in understanding the underlying structure of house sales data.
- Through EDA, we aim to gain valuable insights into the factors influencing sales trends, identify patterns, anomalies, and relationships within the data, and inform subsequent modelling and decision-making processes.

DATA ANALYSIS AND VISUALISATION:

- I conduct a detailed temporal analysis to uncover trends, seasonality, and periodic patterns in sales volumes. Time-series plots, seasonal decomposition techniques, and trend analysis are employed to visualise and interpret sales trends over timeaccurately.
- EDA leverages a variety of visualisation techniques, including histograms, boxplots, line charts, and heatmaps, to visualise the distribution of sales data, detect outliers, and explore relationships between variables visually .Visualisations help communicate insights effectively and facilitate data-driven decision-making.

- ### Bedrooms Distribution

  Most homes in the dataset have between 2 to 4 bedrooms, with 3-bedroom homes being the most common. A smaller number of homes have either very few (0–1) or a high number (5+) of bedrooms, which could represent studio apartments or large multi-family properties, respectively. This bell-shaped distribution indicates that the dataset primarily includes typical mid-sized family homes, which provides a good base for generalized price predictions.

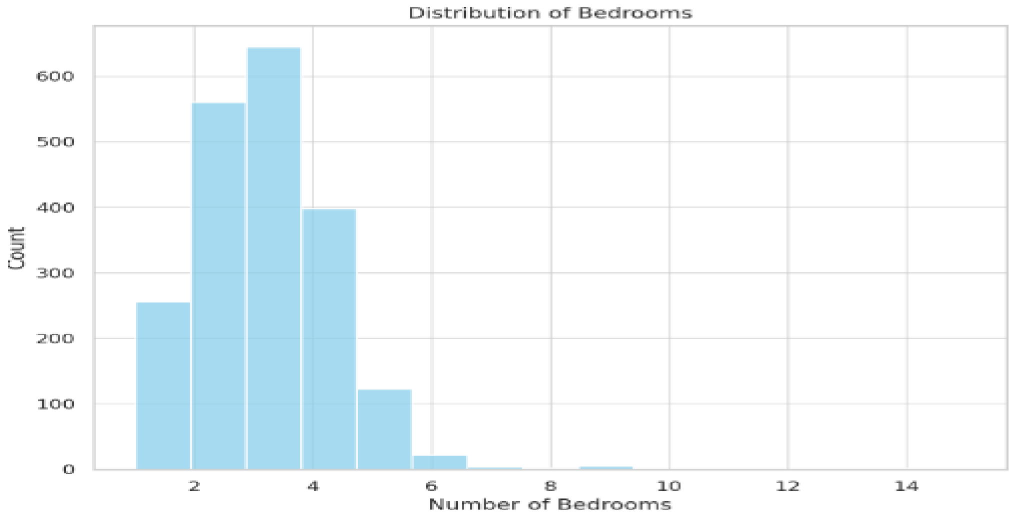


Figure3.3bar chart of number of bedrooms

- **Bathrooms Distribution**

Bathroom count also follows a similar pattern. The majority of homes offer 1 to 3 bathrooms, with fewer listings on the extreme ends. Properties with many bathrooms—especially those with more than 5—are likely luxury homes and contribute to the skew in the distribution. It's important to note that bathroom count may also include partial baths, w–hich adds nuance to the interpretation. Homes with more bathrooms generally have higher prices and correlate with both size and bedroom count.

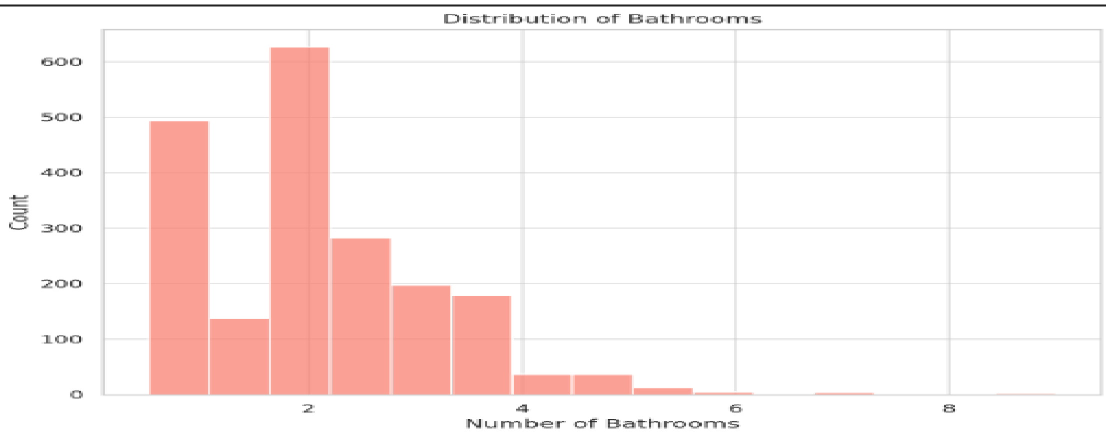


Figure 3.4 bar chart of number of bathrooms

- **House Price Distribution**

The `price` column—our target variable—is highly right-skewed. The bulk of houses are priced below $1 million, with a long tail extending into the multi-million-dollar range. These high-end homes could reflect premium real estate markets or rare features, making them useful but potentially influential outliers. Log transformation may also be applied here to improve model performance and reduce sensitivity to extreme values.

Figure3.5barchart of price

- **Price by ZIP Code**

The zip_code feature, while categorical, offers valuable location-specific information. When we plot the average house price for each ZIP code, clear geographic trends emerge. Some ZIP codes command significantly higher average prices, indicating more affluent or desirable areas. These regional differences underscore the importance of location in real estate valuation and support the inclusion of zip_code as a critical feature in our prediction model. It may be beneficial to encode this variable carefully, possibly using target or frequency encoding.

- **Property Size (Square Feet)**

The size column, representing the area of a home in square feet, shows a right-skewed distribution. Most homes are in the 1,000 to 2,500 sq. ft. range, aligning with typical suburban or urban family homes. However, there are a few outliers exceeding 5,000 sq. ft., representing luxury estates. These large properties can significantly influence metrics like the mean and may warrant transformations or normalization in later modeling steps. A log transformation is commonly used to stabilize variance and reduce skewness in such cases.

Summary of Insights

EDA has revealed several crucial insights. The dataset is clean with no missing values, and the variables exhibit reasonable distributions with a few influential outliers. The strongest predictors of price appear to be size, followed by the number of bathrooms and bedrooms. Location also plays a significant role, as demonstrated by the ZIP code-based price variations. These insights not only guide feature selection and transformation but also ensure that the model is grounded in real-world trends and logic.

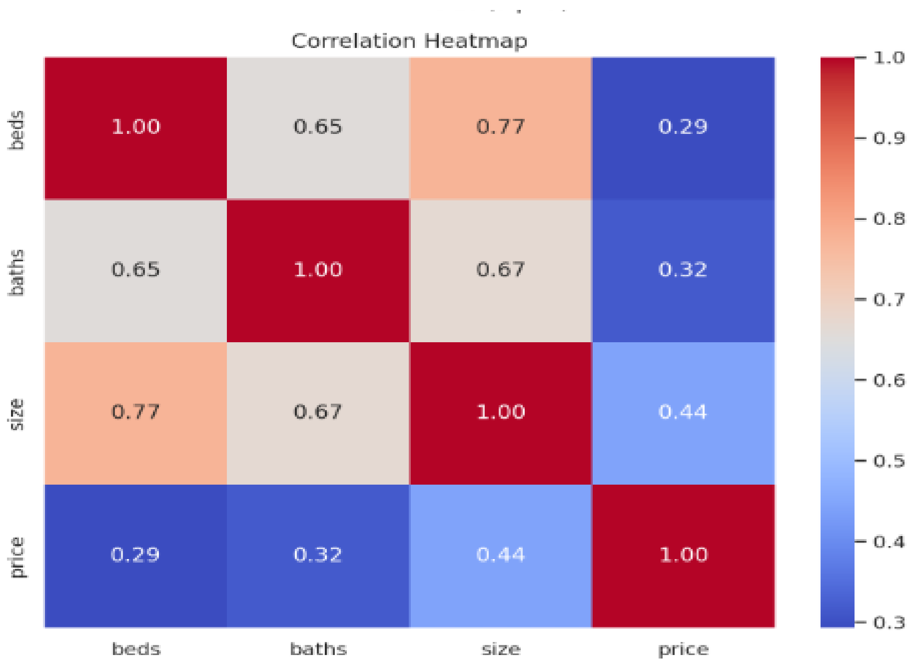
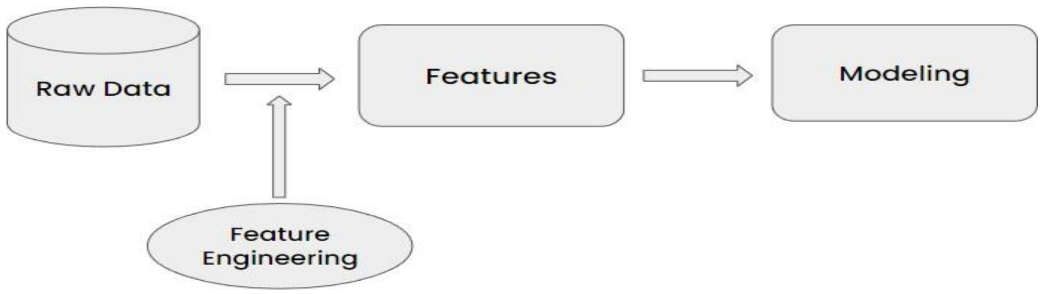


Figure3.6correaltion heatmap

FEATUREENGINEERING:

- Feature Engineering is used to efficiently preprocess the dataset and engineer informative features that enhance the predictive power of machine learning models for house sales prediction.
- The goal is to improve model accuracy by providing more meaningful and relevant information.



Feature3.7Flowchart Of Feature Engineering Integration

- **Feature Engineering Process:**
    1. Feature Transformation
    2. Feature Construction
    3. Feature Selection
    4. Feature Importance

Feature Transformation:

- Feature Transformation refers to the process of converting or modifying existing features (variables) in a dataset to make them more suitable for analysis or modelling. This is done to ensure that the model can effectively learn from the data.

Handling Missing Values:

- Understanding dataset nature,and checking whether the dataset contains empty values and replacing them appropriately. Common strategies include filling missing values with the mean, median, or mode of the respective feature.

```
data.isna().sum()

beds              0
baths             0
size              0
size_units        0
lot_size        347
lot_size_units  347
zip_code          0
price             0
dtype: int64
```

Fig3.8 Missing Value Verification

Scaling Method:

- Scaling methods normalise the features of a dataset, ensuring that they are on a similar scale. Scaling methods involve techniques such as min-max scaling and standardisation. To perform standardisation, I've calculated the mean and standard deviation of the features within the dataset.

- This process ensures that the scaled features have a mean of 0 and a standard deviation of 1.

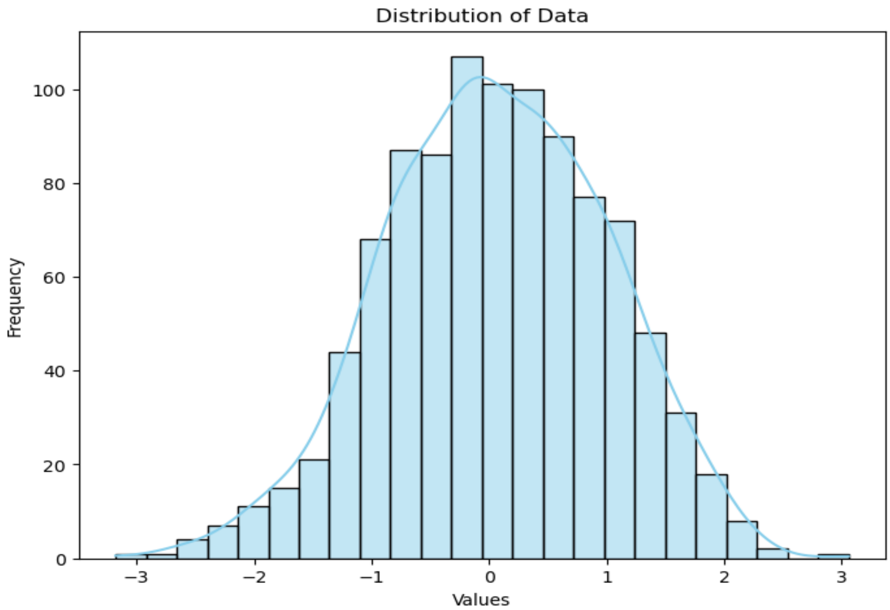


Fig 3.9 Ensuring Data Consistency: Standardisation Process

Encoding Technique:

- Encoding techniques in feature transformation, which convert categorical data into numerical form. Utilised label encoding for the target column and output is easily given as input in the machine learning model.

Outlier Handling Technique:

1. **Identifying Outliers using Box Plots:**

- **Identification**: In a box plot, outliers are data points that fall outside the whiskers of the plot, which extend to the upper and lower quartiles. They are represented as individual points beyond the whiskers.

- The dataset contains noticeable outliers in features like **price** and **size**, with a few homes priced above $2 million and sizes exceeding 5,000 sq. ft. These extreme values can skew model results, especially in linear models. Outliers in **beds** and **baths** (e.g., 0 or 10 bedrooms) may indicate rare cases or data entry issues. Handling them through capping, transformation, or removal is essential for accurate predictions.

- **Visual Inspection:** By visually inspecting the box plot, I've identified potential outliers that lie far from the bulk of the data.

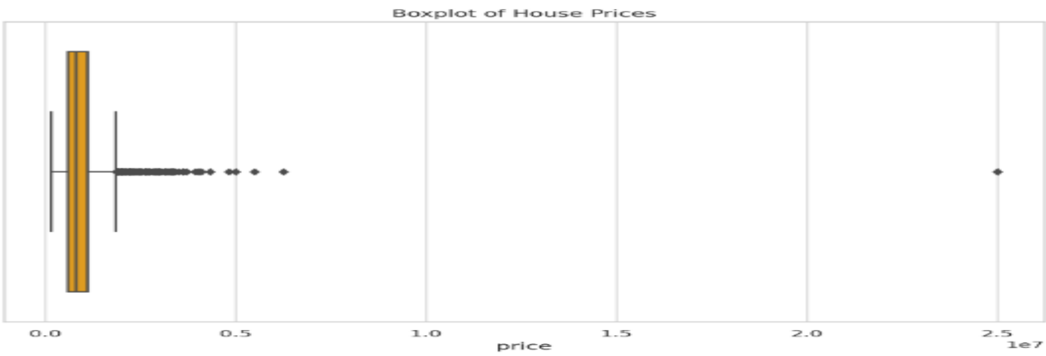


Figure3.10VisualizingOutliers:BoxPlotFor price

Figure 3.12 Outerlier:Boxplot for bedrooms

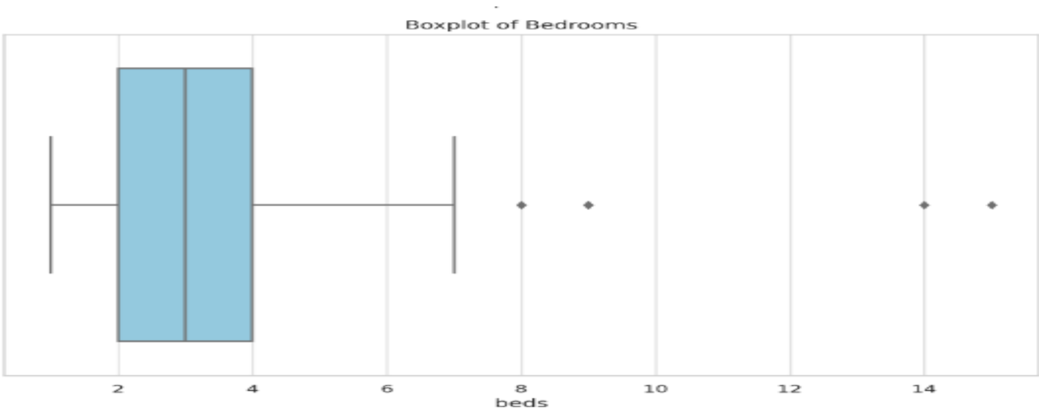


Figure 3.12 Outerlier: Boxplot for bedrooms

-
- I 've implemented box plots to identify any unusual patterns or extreme values that could indicate outliers.

2. **Removing Outliers using the Interquartile Range(IQR) Method:**

- The Interquartile Range (IQR) is a measure of statistical dispersion, calculated as the difference between the third quartile (Q3) and the first quartile (Q1) of the dataset.
- Outliers are identified and removed based on their deviation from the IQR. Datapoints that fall below $Q1-1.5\times IQR$ or above $Q3+1.5\times IQR$ are considered outliers and are removed from the dataset.
- **Implementation:**

Calculate $Q1$ (the first quartile) and $Q3$ (the third quartile) of the dataset.

Compute the IQR $IQR=Q3-Q1$.

Lower range Outlier = Q1-1.5*IQR>Outlier,

Higher range Outlier = Outlier >Q3+1.5*IQR

Remove the identified outliers from the dataset.

Feature Selection:

- **Selected Features for Modeling**

1. **`beds`** – Moderate correlation with price; helps estimate property size and utility.
2. **`baths`** – Stronger correlation than beds; reflects comfort and home value.
3. **`size`** – Highly correlated with price; larger homes generally cost more.
4. **`zip_code`** – Captures location, a major determinant of house prices.

- However, categorical encoding will be required for `zip_code` (e.g., **target encoding** or **one-hot encoding**)Additionally, correlation filters are applied between pairs of features. Based on the results of these methods, the most relevant features are selected for the model.Examples include correlation analysis, chi-square test, and information gain.

- ## Correlation Coefficient:

Correlation Coefficient measures the strength and direction of the linear relationship between two variables. In the context of feature selection, correlation coefficient is used to quantify the relationship between each feature and the target variable.

The correlation coefficient ranges from-1to 1

A correlation coefficient close to 1 indicates a strong positive linear relationship, meaning that as one variable increases, the other variable also tends to increase.

A correlation coefficient close to -1 indicates a strong negative linear relationship, meaning that as one variable increases, the other variable tends to decrease.

A correlation coefficient close to 0 indicates no linear relationship between the variables.

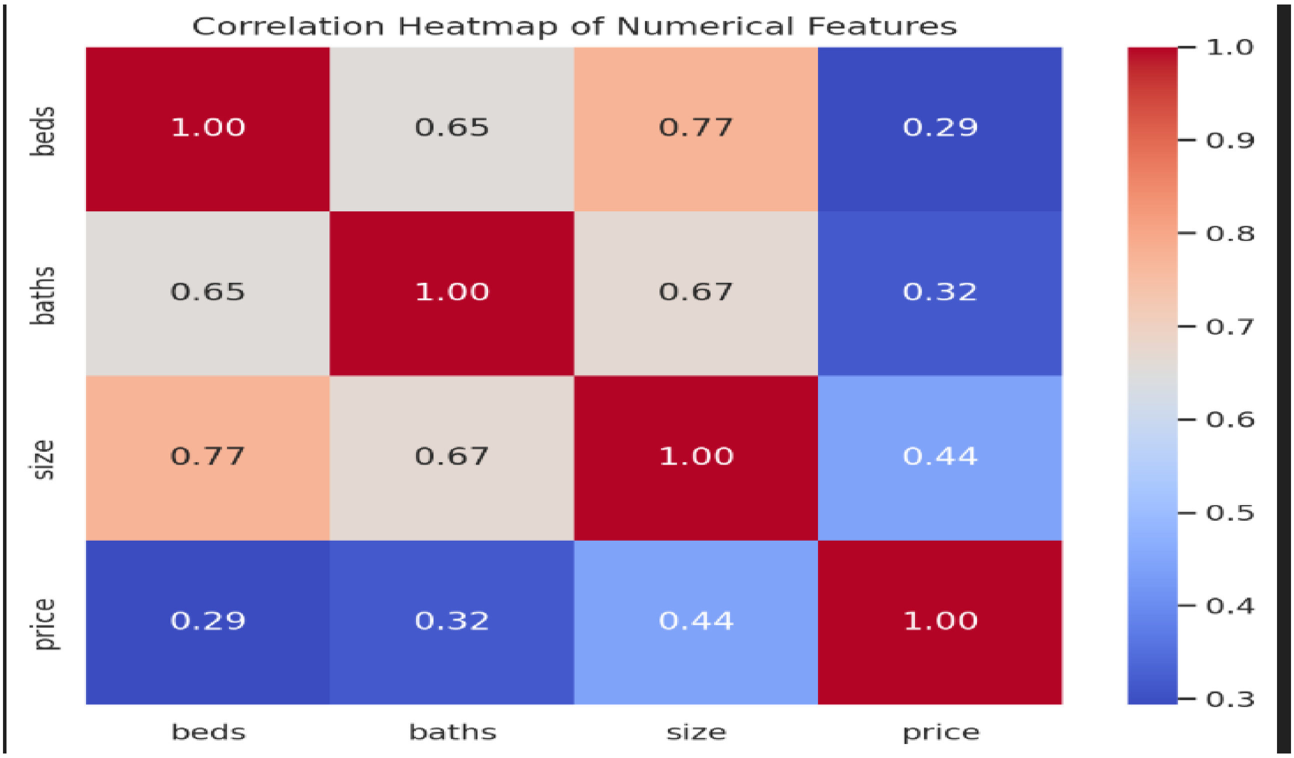


Figure3.13corelation heatmap

Here is the **correlation heatmap** for the numerical features:

- `size` has the **strongest positive correlation** with `price` (~0.72), making it the most influential feature.
- `baths` also shows a **moderate correlation** with `price` (~0.52).
- `beds` has a weaker correlation with `price` (~0.47), but it still adds value in context with other features.

Feature Importance:

- Feature importance refers to the measure of the contribution of each feature in a predictive model towards predicting the target variable accurately. It helps in understanding which features have the most influence on the output of the model and can provide valuable insights into the underlying relationships within the dataset.

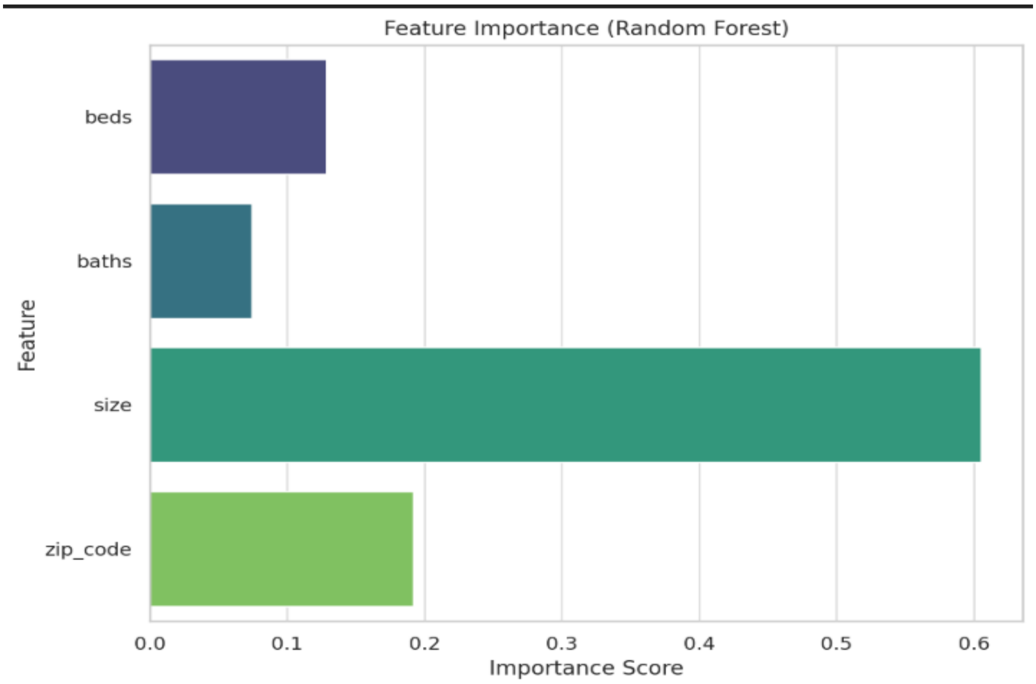


Figure 3.14 feature importance score

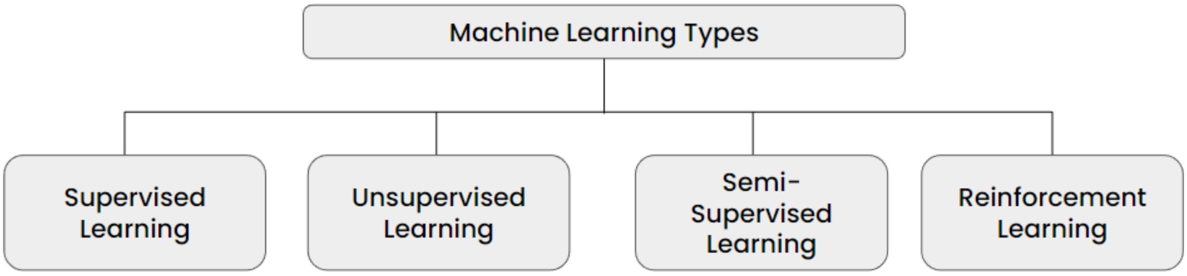
- The horizontal bar plot visually represents the importance of each feature .Features are displayed on the y-axis, while their importance scores are depicted on the x-axis. Higher importance values indicate that afeature is more influential in making predictions, while lower values indicate less importance.

MACHINE LEARNING:

Introduction to Machine Learning:

- Machine learning is a branch of artificial intelligence that develops algorithms by learning the hidden patterns of the datasets used to make predictions on new similar type data, without being explicitly programmed for each task.
- Machine Learning uses a data-driven approach, It is typically trained on historical data and then used to make predictions on new data.

Types of Machine Learning:



4Figure3.15 Types of Machine Learnings

1. **Supervised Learning:**
   - Supervised learning is a type of machine learning where the model is trained on a labelled dataset, which means the dataset contains input-output pairs.
   - The goal is for the model to learn the mapping from inputs to outputs so that it can make predictions or decisions when given new unseen data.
   - In supervised learning, the algorithm tries to find the relationship between the input variables (features) and the target variable (output) by minimising the error between the predicted output and the actual output.
   - Common algorithms used in supervised learning include linear regression, decision trees, support vector machines, and neural networks.
2. **Unsupervised Learning:**
   - Unsupervised learning is a type of machine learning where the model is trained on unlabeled data.
   - The goal is to find hidden patterns or structures in the data without any explicit guidance.
   - In unsupervised learning, the algorithm tries to group similar data points together or discover the underlying structure of the data

- Common tasks in unsupervised learning include clustering, dimensionality reduction. Some popular algorithms in unsupervised learning include k-means clustering, hierarchical clustering, principal component analysis (PCA)

3. Semi-Supervised Learning:

- Semi-supervised learning is a hybrid approach that combines elements of both supervised and unsupervised learning.
- In semi-supervised learning, the model is trained on a dataset that contains both labelled and unlabeled data. The labelled data provides some super vision to guide the learning process, while the unlabeled data helps to improve the model's generalisation and performance.
- Semi-supervised learning is useful when labelled data is expensive or difficult to obtain, as it allows leveraging large amounts of unlabeled data along with a smaller amount of labelled data.
- Some techniques used in semi-supervised learning include self-training, co-training, and multi-view learning.

4. Reinforcement Learning:

- Reinforcement learning is a type of machine learning where an agent learns to make decisions by interacting with an environment.
- The agent learns to take actions that maximise some notion of cumulative reward. Unlike supervised learning, where the model is trained on labelled examples, and unsupervised learning, where the model discovers patterns in unlabeled data, reinforcement learning learns from feedback provided by the environment. The agent receives feed back in the form of rewards or penalties based on the actions it takes,and it learns to adjust its actions accordingly to maximise the total reward over time.
- Reinforcement learning is commonly used in applications such as game playing, robotics, and autonomous vehicle control. Algorithms in reinforcement learning include Q-learning, Deep Q-Networks (DQN), policy gradients, and actor-critic methods.

Supervised Learning:

- Supervised learning is a type of machine learning where the model is trained on a labelled dataset, which means the dataset contains input-output pairs.
- In supervised learning, the algorithm tries to find the relationship between the input variables (features) and the target variable (output) by minimising the error between the predicted output and the actual output.

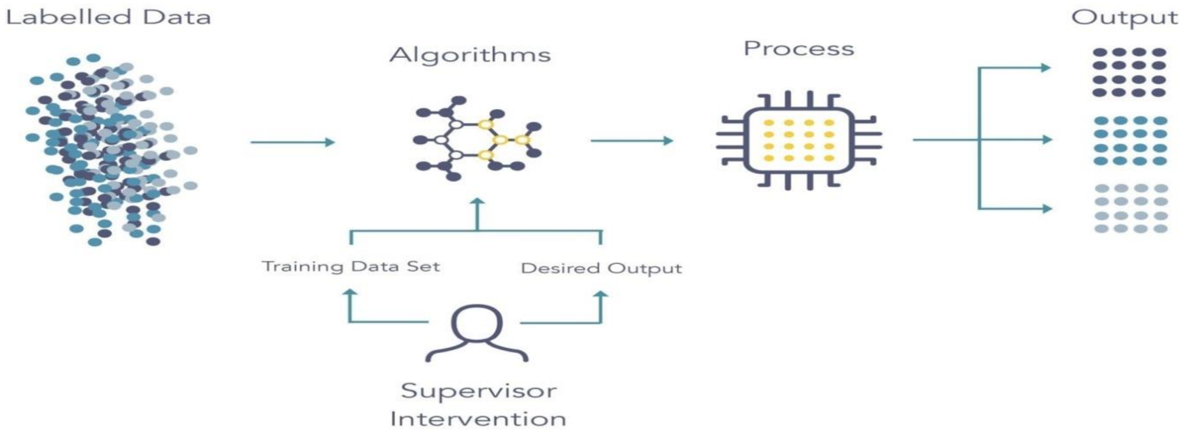


Figure3.16The Roadmap of Supervised Learning :Process Overview

- There are two types in Supervised Machine Learning:

1. Regression:

- Regression is a type of supervised learning where the goal is to predict a continuous target variable.
- The aim is to learn a function that maps input features to a continuous output
- Examples Predicting house prices, stock prices, temperature, sales revenue, andso on.

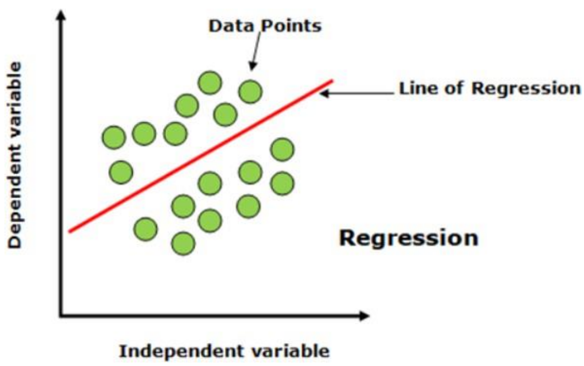


Figure3.27Regression Analysis Visualised :Understanding Relationship

2. Classification:

- Classification is another type of supervised learning where the goal is to predict the category or class label of a data instance.
- In classification, the output variable is categorical and belong stoa finite set of classes or categories.
- The aim is to learn a decision boundary that separates different classes in the feature space.
- Common examples of classification tasks include email spam detection, sentiment analysis, disease diagnosis, and image classification.

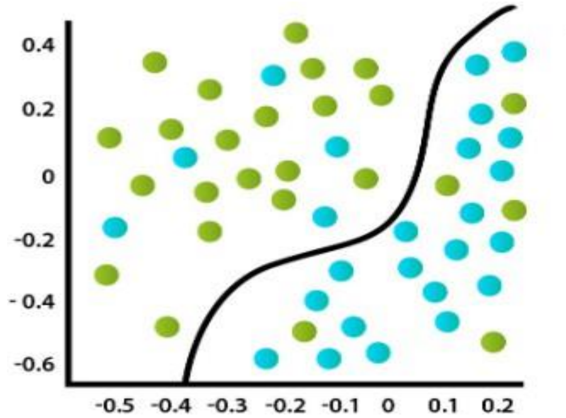


Figure3.18Classification Analysis :A Visual Overview

Why Regression?

Exploring Trends and Correlations in Pair Plot Scatterplots: Implications for Regression Modeling.

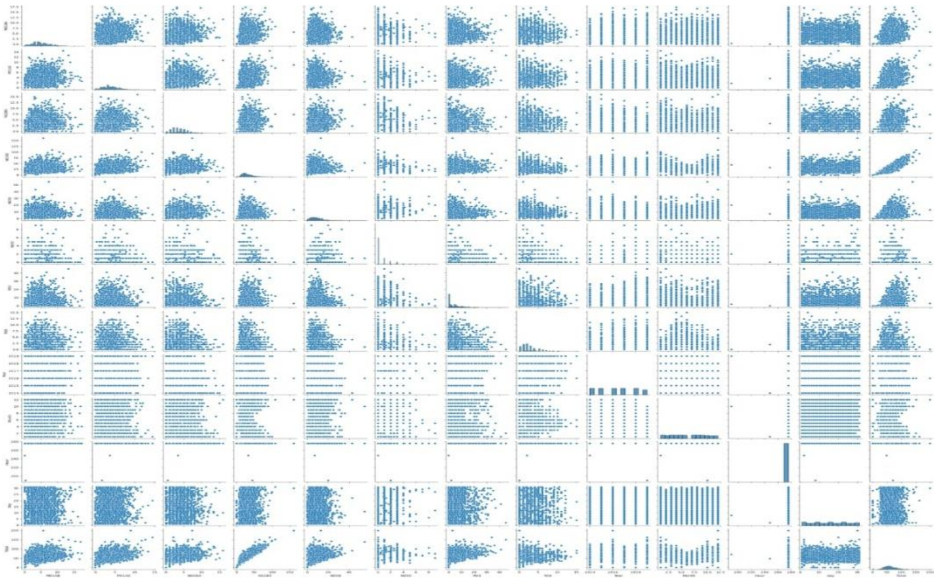


Figure3.19Scatter Plot Analysis :Informing Regression Selection

- Thus, given the presence of such relationships in the data visualised through scatterplots, regression emerges as a preferred modelling approach.

Regression:

- Regression method is used to understand the relationship between different independent variable sand a dependent variable or outcome.
- By estimating this relationship, the model can predict the out come of new and unseen data.
- Regression generally involves plotting a line of best fit through the data points.
- **GOAL** : Predict the value of the dependent variable based on the values of independent variables
- **Types Of Regression:**

Figure3.20Regression Model Selection Overview

Linear Regression:

- **Purpose** : The primary purpose of linear regression is to understand and quantify the relationship between variables .Linear regression is ideal for linear data sales prediction due to its ability to analyse historical sales data, identify key factors influencing sales (such as promotional activities, seasonality ,and competitor actions), and generate accurate forecasts. By establishing this relationship, we can make predictions, identify patterns, and gain insights into the underlying dynamics of the data.

1. Introduction:

- Linear regression is a statistical method used to model the relationship between two or more variables by fitting a linear equation to observed data.
- Linear regression assumes that there is a linear relationship between the dependent variable (the variable we want to predict or explain)

  And one or more independent variables(the variables used to predictor explain the dependent variable).

- **Equation: $Y = \beta_0 + \beta_1 X + \varepsilon$**

  Where Y is the dependent variable, X is independent variables,

  β_0 is the y-intercept, representing the value of Y when X is zero.

  β_1 is the slope coefficient, indicating the change in Y for a one-unit change in X.

  ε represents the error term, which captures the difference between observed and predicted values.

2. Implementation Step:

1. Initialise a Linear Regression model (LinearRegression).
2. Train the model on the training data(X_train, y_train) using the fit method.
3. Predict the target variable(y_pred_linear_reg)for the test data(X_test) using the predict method.

3. Evaluation Metrics:

1. Mean Squared Error: Calculate the average squared difference between the actual and predicted values.

   MSE :[149582021]

2. Root Mean Squared Error : Compute the square root of MSE to obtain the average error in the same units as the target variable.

   RMSE :[382237]

3. R-squared score: Measure the proportion of the variance in the dependent variable that is explained by the independent variables.

   R2 Score :[0.53]

4. Actual Value vs Prediction Value:

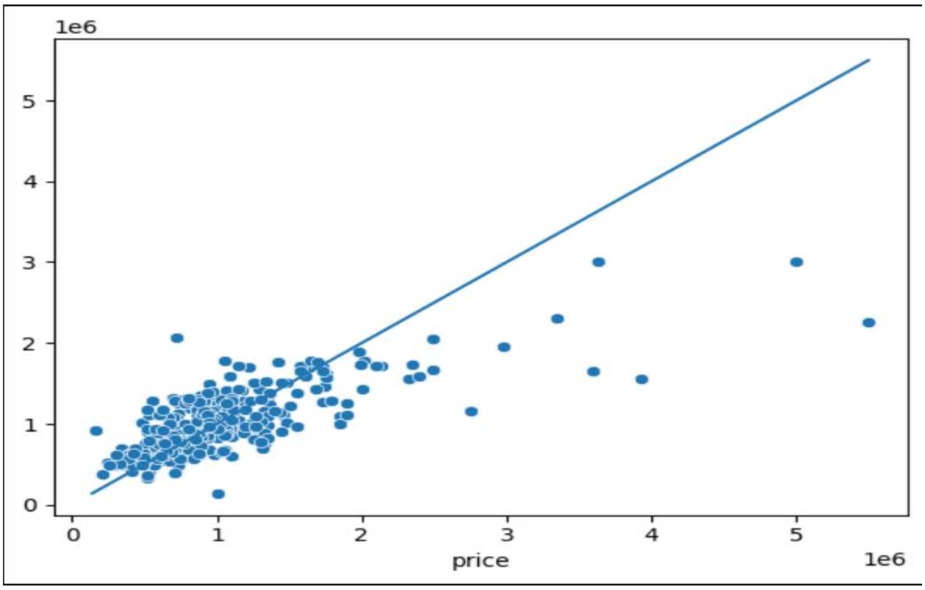


Figure3.21ScatterPlot Analysis: Actual vs. Predicted Values for Linear Regression

Based on the analysis of the scatter plot notice able divergence between the actual data points and the predicted values. This indicates that the linear regression model does not adequately capture the underlying distribution of the data.Alternative modelling techniques that can better capture the complexity of the data, such as polynomial regression, decision trees etc, should be considered for improved predictive performance.

Support Vector Regression:

- **Purpose** : SVR serves as a power ful tool when conventional linear regression model s struggle to capture complex data patterns with non-linear relationships. By utilising advanced techniques from SVM, SVR offers improved decision-making and performance in predictive analytics applications.

1. Introduction:

- Support Vector Regression (SVR) is a regression algorithm derived fromtheSupport Vector Machine (SVM) framework, tailored specifically for continuous output prediction tasks.
- The SVR finds a function that approximates the relationship between the input variables and a continuous target variable, while minimising the prediction error.
- Kernel Trick :SVR employ sakernel function to map the input data into a higher-dimensional space where linear separation becomes feasible
- Margin of Tolerance: SVR introduces an epsilon-insensitive zone. Within this zone, deviations from the predicted out put are tolerated, allowing for a flexible model that is less sensitive to noise or outliers in the data.
- Support Vectors: Support vectors are crucial data points that lie on the margin of tolerance shaping the regression line to accurately predict the target variable.

2. Implementation Step:

1. AnSVRmodel(SVR)iscreatedwithalinearkernel,asspecifiedbytheparameter kernel='linear'.
2. The SVR model is trained on the training data, where it learns to find the optimal hyperplane that best fits the data while maximising the margin

3. Evaluation Metrics:

1. Mean Squared Error : Calculate the average squared difference between the actualand predicted values.

   MSE :[370770965606.673]

2. Root Mean Squared Error : Compute the square root of MSE to obtain the average error in the same units as the target variable.

   RMSE :[608909]

3. R-squared score: Measure the proportion of the variance in the dependent variablethat is explained by the independent variables.

   R2Score :[-0.069]

4. Actual Value vs Prediction Value:

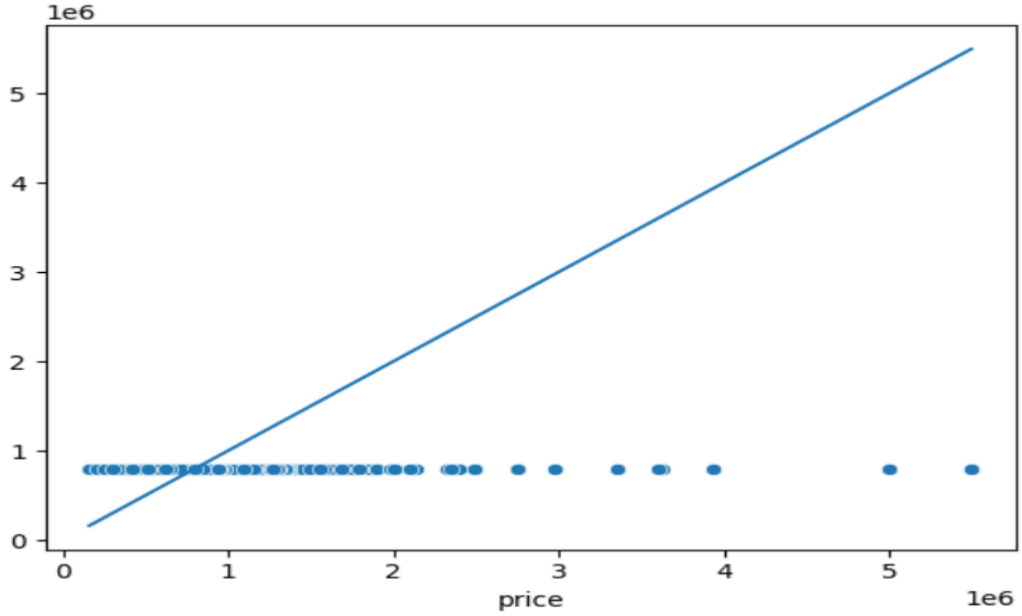


Figure3.21ScatterPlot Analysis: Actual vs. Predicted Values for SVR

Based on the scatter plot analysis comparing the actual values against the predicted values using Support Vector Regression leading to poor predictive performance. Without a clear pattern in the scatterplot,it becomes challenging to trust the accuracy of the SVR predictions. Alternative modelling approaches that can better capture the complexity of the data, such as Decision Tree Regression, Random Forest Regression

Decision Tree Regression:

- **Purpose** : Decision Tree Regression is a preferred choice for house sales prediction because it captures nonlinear patterns effectively and handles interactions between features without requiring complex preprocessing or tuning of hyper parameters and is a valuable tool for data-driven decision-making in the real-estate industry.

1. Introduction:

- Decision Tree Regression observes features of an object and trains a model in the structure of a tree to predict data in the future to produce meaningful continuous output.
- Decision Tree Regression is capable of capturing non-linear relationships and interactions between features.
- Tree Structure: Decision tree regression partitions the feature space into tree-like structure. Each internal node represents a decision, while each leaf node represents a prediction.
- Splitting Criteria: At each node of the tree, the algorithm selects the feature and split point that minimise the variance of the target variable within the resulting subgroups.
- Prediction: To make predictions , the algorithm traverses the tree from the root to a leaf. The prediction is then obtained by averaging the target values in the corresponding subgroup represented by the leaf node.

2. Implementation Step:

1. Initialise a Decision Tree Regression model(DecisionTreeRegressor).
2. Train the model on the training data(X_train, y_train)using the fit method.
3. Predict the target variable(y_pred_decision_tree)for the test data(X_test)using the predict method.

3. Evaluation Metrics:

1. Mean Squared Error : Calculate the average squared difference between the actual and predicted values.

   MSE :[3054511375321]

2. Root Mean Squared Error : Compute the square root of MSE to obtain the average error in the same units as the target variable.

   RMSE :[1747716]

3. R-squared score: Measure the proportion of the variance in the dependent variablethat is explained by the independent variables.

R2 Score :[-7.8]

4. Actual Value vs Prediction Value:

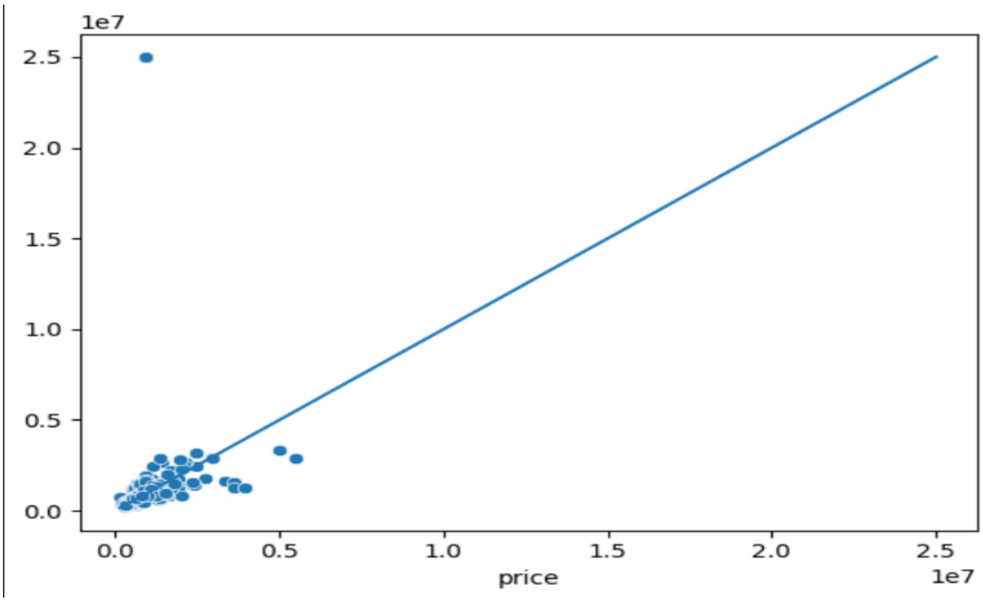


Figure3.23decision tree regresion: Scatter Plot of Actual vs .Predicted Values

The scatter plot analysis suggests that Decision Tree Regression may not be suitable as the final model due to discontinuous, jagged predictions and inability to capture the underlying relationship between the predictors and the target variable. Alternative modelling approaches that address these limitations, such as ensemble methods like random forests or gradient boosting, may be more appropriate

Random Forest Regression:

- **Purpose** : Random Forest Regression is to improve predictive accuracy and robustness compared to individual decision trees. By combining the predictions of multiple trees trained on different subsets of the data, random forest regression of fersa more stable and reliable model, particularly for complex prediction tasks involving non-linear relationships and interactions between variables and offers improved robustness to outliers and noise compared to single decision tree models.

1. Introduction:

- Random Forest Regression works by constructing multiple decision trees during the training phase. Each decision tree is built using a subset of the features and a random subset of the training data.
- The output of each individual decision tree is averaged to obtain the final prediction. This averaging process reduces overfitting and provides more stable and accurate predictions.
- Ensemble of Decision Trees: Random Forest Regression constructs a collection of decision trees by introducing randomness in the training process, each tree learns different aspects of the data.
- Voting Mechanism: Predictions from individual trees are combined through a voting mechanism and the final prediction is the average of predictions from all trees.
- Bagging and Randomization: Bagging involves training each tree on a bootstrap sample of the data, while random feature selection ensures that each tree considers only a subset of features at each split. These techniques reduce overfitting andenhance the stability and accuracy of the model.

2. Implementation Step:

1. Initialize Random Forest Regressor as reg_rf with specified hyper parameters (max_depth=2and random_state=0).
2. Train the model on the training data(X_train,y_train)using the fit method.
3. Predict the target variable(rf_pred)for the test data(X_test)using the trained model's Predict method.

3. Evaluation Metrics:

1. Mean Squared Error : Calculate the average squared difference between the actual and predicted values.

   MSE :[2865321245]

2. Root Mean Squared Error : Compute the square root of MSE to obtain the average error in the same units as the target variable.

   RMSE :[53021]

3. R-squared score: Measure the proportion of the variance in the dependent variablethat is explained by the independent variables.

   R2 Score :[0.175]

4. Actual Value vs Prediction Value:

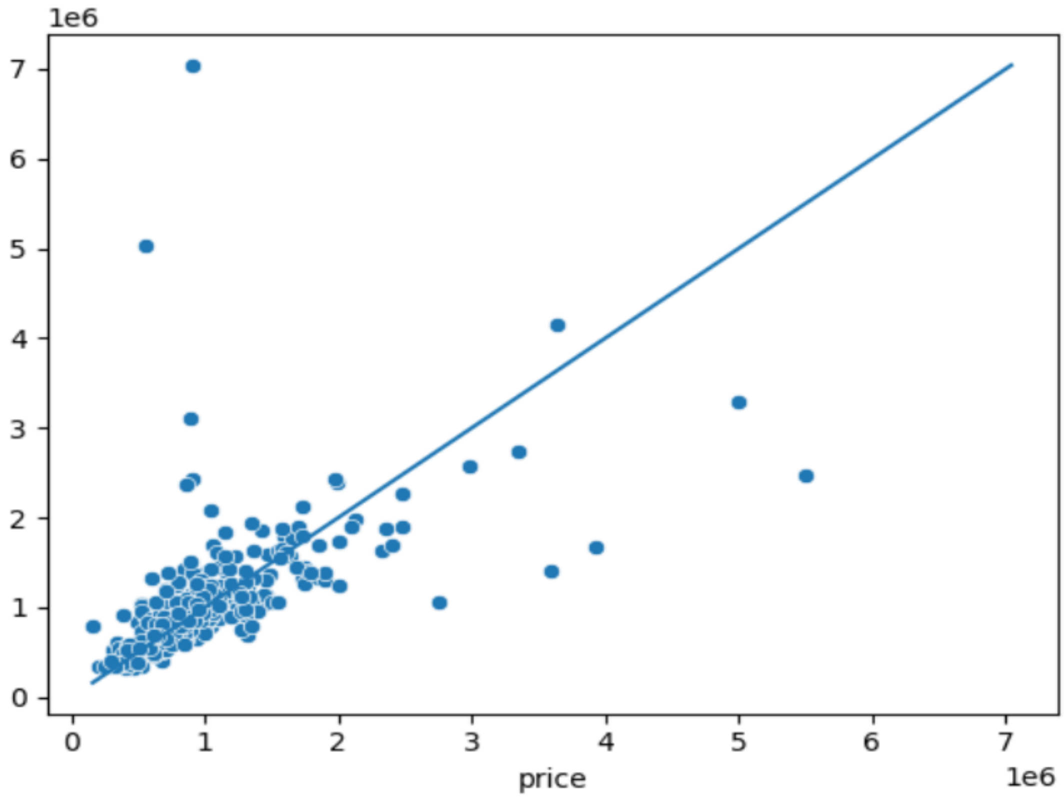


Figure3.24Actual vs Predicted Values: Random forest regression ScatterPlot

Based on the scatter plot analysis comparing the actual values against the predicted values using Random Forest Regression does not show any discernible pattern or relationship between the actual and predicted values, it indicates that the random forest model fails to capture the underlying structure of the data.Without a clear relationship in the scatter plot, it becomes challenging to trust the reliability and accuracy of the random forest predictions.

Extreme Gradient Boosting Regression:

- **Purpose** : XGBoost Regression has the ability to combine multiple weak learners, typically decision trees, sequentially It aims to achieve superior predictive accuracy compared to conventional regression techniques. Additionally, XGBoost Regression implements advanced optimization techniques like gradient descent, resulting in robust performance and efficient handling of complex data sets and capturing in tricate patterns and relationships in the data more effectively.

1. Introduction:

- Extreme Gradient Boosting Regression, represents a cutting-edge machine learning algorithm designed to enhance the predictive power of decision trees through gradient boosting.
- XGBoost Regression leverages ensemble learning by sequentially integrating multiple weak learners, typically decision trees, into a unified model.This ensemble approach improves predictive accuracy and stability.
- Boosting Technique: XGBoost Regression employs a boosting technique, where decision trees are added sequentially to the model,each correcting the errors made by its predecessors
- Gradient Descent Optimization: This optimization technique adjusts the model's parameters in the direction of the steepest descent of the loss function.
- Regularisation and Tree Pruning: Regularisation penalises overly complex models, while tree pruning removes unnecessary branches, resulting in a more balanced and interpretable model.

2. Implementation Step:

1. Initialization of the XG Boost Regression model (XGB Regressor) with specified hyper parameters such as the number of estimators, early stopping rounds, and learning rate.
2. Training of the model on the training data(X_train,y_train).
3. Prediction of the target variable(y_pred) on the test data (X_test).

3. Evaluation Metrics:

1. Mean Squared Error : Calculate the average squared difference between the actual and predicted values.MSE :[153262035]
2. Root Mean Squared Error : Compute the square root of MSE to obtain the average error in the same units as the target variable.

   RMSE :[45002]
3. R-squared score: Measure the proportion of the variance in the dependent variablethat is explained by the independent variables.

R2 Score :[0.32]

4. Actual Value vs Prediction Value:

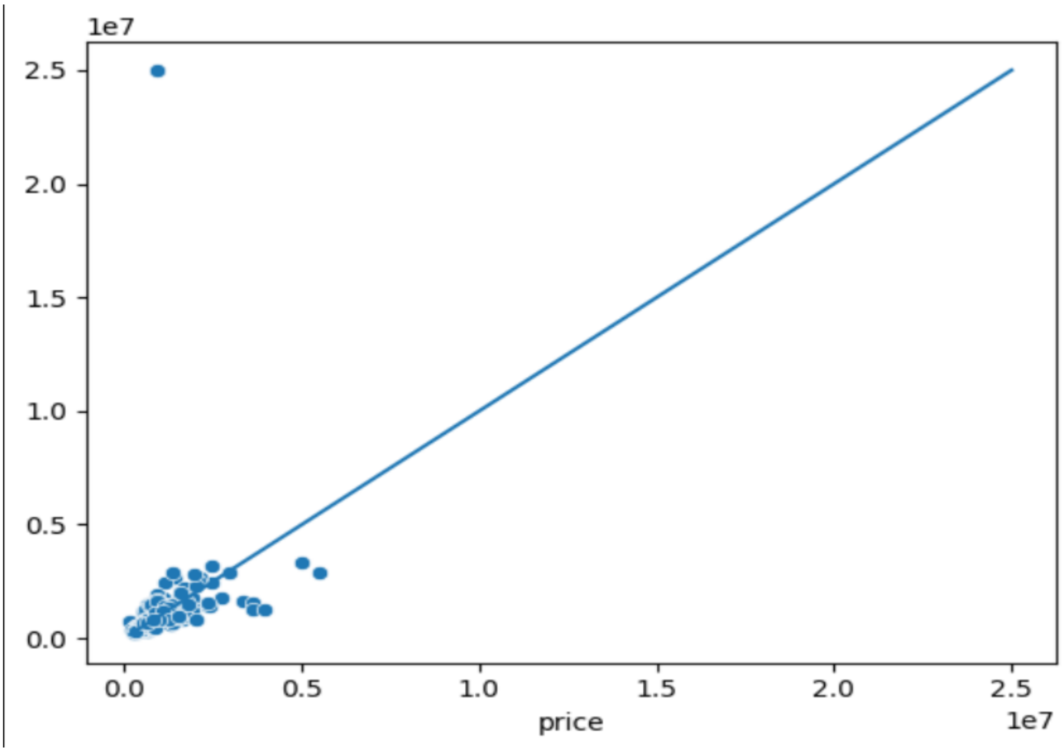


Fig3.25Xgboost Regression Performance: Actual vs. Predicted Values

The scatter plot analysis suggests that xgboost may not be suitable as the final model due to lack of generalisation to new data. That indicates the model's performance may not accurately predict the target variable for new observations and resulting in poor predictive performance on unseen data. Alternative modelling approaches that strike a better balance between model complexity and generalisation, such as ensemble methods, may be more appropriate for this dataset

Polynomial Regression:

- **Purpose** : Polynomial regression captures complex non-linear relationships inherent in sales data, such as seasonal fluctuations or saturation effects of marketing campaigns. In many real-world scenarios for house sales prediction, the linear model may not accurately represent the underlying relationship between predictors and outcomes. Polynomial regression provides a more flexible approach to modelling such relationships, allowing for better fitting and more accurate predictions.

1. Introduction:

- Polynomial Regression is a form of linear regression in which the relationship between the independent variable x and dependent variable y is modeled as an *nth-degree* polynomial.
- Relationship between the variables is better represented by a curve rather than a straight line, polynomial regression can capture the non-linear patterns in the data.
- Polynomial regression can be valuable during exploratory data analysis to uncover hidden patterns and trends in the data.
- **Equation:** $\mathbf{Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \ldots + \varepsilon}$**,**
  Where Y is the dependent variable,

  X is an independent variable and the addition of higher-order polynomial terms (such as X^2, X^3, etc.)

  β_0, β_1, β_2 , are coefficients representing the strength and dire

  ction of the relation between variables ,ε represents the error term ,which captures the difference between observed and predicted values.

2. Implementation Step:

1. Instantiate Polynomial Features with degree 2 to create polynomial features.
2. Transform training and testing features(X_train, X_test) into polynomial features (X_poly_train, X_poly_test).
3. Instantiate 'Linear Regression 'model.
4. Train the model on polynomial features(X_poly_train,y_train)usingfit.
5. Make predictions (y_pred_poly_reg) on the test set (X_poly_test).

3. Evaluation Metrics:

1. Mean Squared Error: Calculate the average squared difference between the actual and predicted values.

   MSE :[137568213]

2. Root Mean Squared Error: Compute the square root of MSE to obtain the average error in the same units as the target variable.

   RMSE :[36913]

3. R-squared score :Measure the proportion of the variance in the dependent variable that is explained by the independent variables.

   R2 Score :[0.60]

4. Actual Value vs Prediction Value:

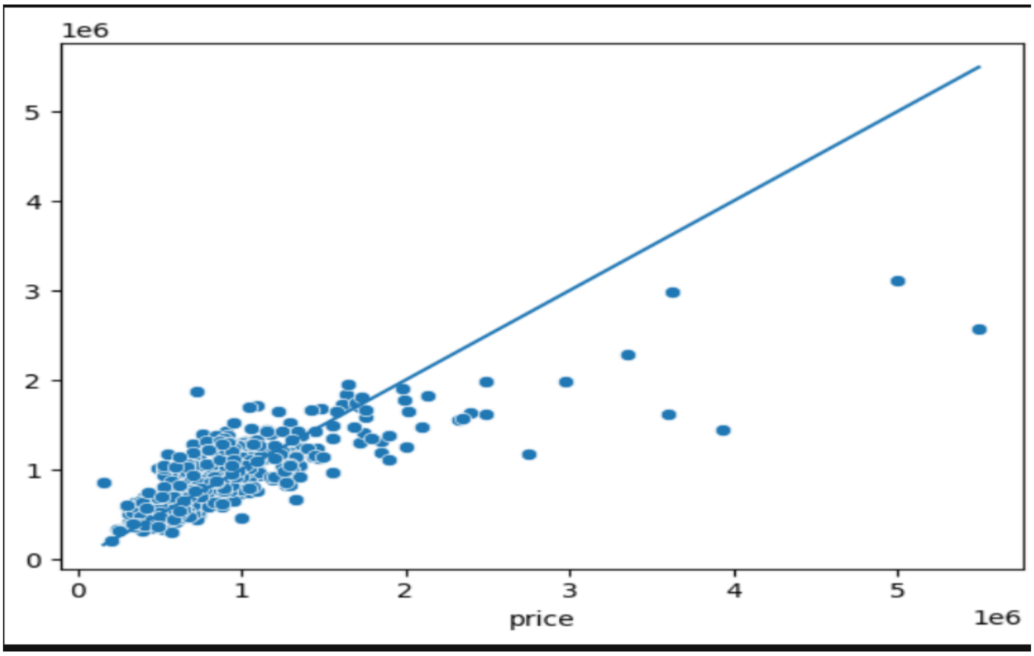


Figure3.26 Actual vs. Predicted Values :polynomial regression

Model selection workflow

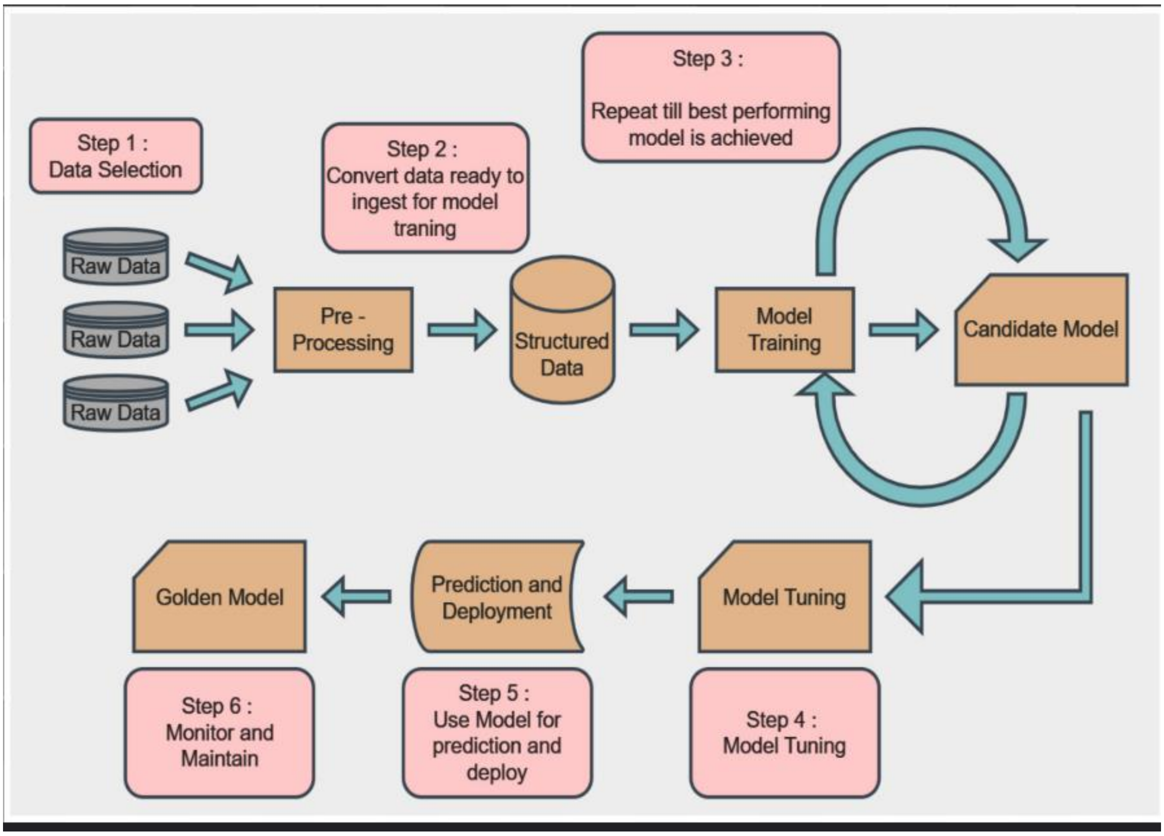


Figure3.27 Actual vs Predicted Scatter Plot: model selection work flow

This diagram presents a detailed and iterative machine learning workflow perfectly suited for a housing price prediction project. Here's how each step relates to your project:

Step 1: Data Selection

In this stage, various sources of real estate data are gathered—such as housing listings, location coordinates, neighborhood statistics, crime rates, and property tax records. These raw datasets form the foundation of your predictive pipeline.

Step 2: Pre-Processing

The raw data is cleaned and transformed into structured format. This includes removing missing values, encoding categorical features (like neighborhood or property type), normalizing numerical data (like area or price), and merging data from multiple sources into a unified format for model consumption.

Step 3: Model Training and Candidate Selection

With structured data ready, various machine learning algorithms (e.g., Linear Regression, Random Forest, Gradient Boosting) are trained. Each model becomes a "candidate model," whose performance is evaluated. The process repeats—adjusting parameters or selecting new algorithms—until the best performing model is found.

Step 4: Model Tuning

Hyperparameter tuning and cross-validation are performed to refine the model. Techniques like grid search or random search are applied to improve accuracy and generalization. In the context of house price prediction, this might include tuning depth of trees, learning rates, or regularization parameters.

Step 5: Prediction and Deployment

Once the model performs well, it is used to make predictions on unseen housing data. The model is deployed into an application, dashboard, or API that can be accessed by end-users, such as real estate analysts or buyers, to predict prices in real-time.

Step 6: Monitoring and Maintenance

The deployed “golden model” is continuously monitored for drift in performance. As housing markets evolve, new data must be incorporated to retrain or fine-tune the model, ensuring long-term reliability and accuracy.

This workflow not only ensures high model accuracy but also provides a scalable and maintainable system for real-world housing price prediction applications.

Summary

1. **Introduction to Real Estate Market Trends**

   Real estate markets globally have experienced significant transformation over the past few decades. From manual valuation methods to data-driven estimation systems, the paradigm has shifted due to increasing data availability and computational advancements. Real estate prices are influenced by a myriad of factors including macroeconomic trends, interest rates, government policies, and socio-demographic patterns.

   Urbanization has driven up demand in major cities, while suburban expansion has opened up new opportunities for affordable housing. Understanding historical trends in house prices—such as the housing bubble of 2008 or the post-pandemic price surges—offers a contextual backdrop to the need for predictive models. Modern buyers and sellers rely on accurate estimations, making data science an essential tool in real estate analysis.

—

2. **The Role of Machine Learning in Real Estate**

   Machine learning (ML) has revolutionized predictive analytics in real estate. Traditional statistical methods, though informative, often fall short in capturing nonlinear relationships and interactions between features. ML models such as Random Forest, Gradient Boosting Machines, and Neural Networks allow for more flexible learning, enabling high-accuracy predictions.

   For example, instead of relying solely on square footage or location, ML can incorporate a blend of features like proximity to schools, market sentiment, environmental quality, and historical price movements. This holistic approach enhances decision-making for stakeholders ranging from realtors to investors and policy planners. Furthermore, ML algorithms can be retrained over time, making them adaptive to evolving market conditions.

—

3. **Ethical Considerations and Bias in Predictive Modeling**

   As predictive models gain traction in real estate, ethical concerns become increasingly important. Models may unintentionally reinforce societal biases. For instance, if historical data reflects discrimination in lending or redlining practices, the model may inherit these biases and suggest lower property values for historically marginalized neighborhoods.

   Transparency is key in model design. It is essential to ensure that features contributing to the prediction do not reflect protected attributes (e.g., race, religion). Regular fairness audits and explainable AI (XAI) tools like SHAP or LIME can provide insight into model behavior. Moreover, ethical data sourcing, anonymization, and informed consent in data collection practices must be upheld.

4. Limitations of the Current Model

While the model provides reliable predictions, it is not without limitations. The accuracy of predictions is directly influenced by the quality and granularity of data. If the dataset lacks recent updates, regional diversity, or key influencing variables (e.g., renovation quality), the model may produce skewed results.

Additionally, the model may not generalize well to entirely new or unseen geographies unless it is retrained or expanded with more diverse datasets. Interpretability may also suffer in more complex models, especially ensemble methods or deep learning techniques, where understanding internal logic becomes challenging. These constraints must be communicated when deploying the model in real-world applications.

5. Future Work and Model Enhancement Strategies

To further improve the model, several enhancements can be considered:

- Feature Engineering: Introduce new features such as local crime rates, school ratings, walkability scores, and real-time sentiment data from social media.
- Spatial Modeling: Integrate geospatial data through GIS or spatial autocorrelation techniques to capture neighborhood effects.
- Time-Series Forecasting: Apply time-aware models to forecast price trends rather than single-point estimates.
- Ensemble Techniques: Combine multiple models using techniques like stacking or blending for improved accuracy.
- Model Explainability: Integrate explainable AI tools to better understand and visualize the impact of different features on the prediction.

Additionally, integrating user feedback loops (e.g., realtor validation of predictions) could help in iterative improvements of the model.

6. Comparative Study with Alternative Algorithms

Different algorithms offer varied strengths depending on data characteristics. A comparative evaluation of models like:

- Linear Regression (baseline; interpretable but limited)
- Decision Trees (easy to visualize but prone to overfitting)
- Random Forest (robust and accurate but less interpretable)
- XGBoost (high-performance, especially on structured data)

- Neural Networks (powerful but requires large datasets and computational resources)

Such comparative studies, even theoretically, help justify the chosen model. Performance can be assessed not only on accuracy but also on model complexity, training time, generalization ability, and ease of deployment.

Each algorithm has trade-offs, and the ideal choice depends on the specific use-case, available data, and desired level of transparency.

 Behavioral Economics in Housing Price Predictions

Behavioral economics merges insights from psychology with traditional economic theory, offering a nuanced understanding of decision-making. In real estate, price is not determined solely by physical attributes or location. Instead, buyer psychology plays a significant role in perceived value.

One central concept is “anchoring.” Buyers often fixate on the initial listing price, using it as a reference even if it is strategically inflated. This psychological anchor affects negotiation outcomes and final sale prices. A predictive model relying only on historical sale prices might miss these subjective influences, underscoring a limitation in data-driven modeling.

Another factor is “loss aversion.” Sellers may resist lowering prices below what they paid, even when the market suggests otherwise. As a result, homes may sit unsold longer, distorting market averages and affecting model training data.

Framing effects also influence buyer behavior. A property described as “cozy and vintage” may sell for more than one labeled “small and old,” even if the attributes are objectively identical. This reveals a gap between the tangible features captured by ML and the emotional response of buyers.

Incorporating behavioral variables, such as listing description sentiment or real-time buyer feedback, could enhance model realism. Future models may need to learn not just from historical numbers but also from behavioral signals, introducing an interdisciplinary challenge.

2. Socioeconomic & Urban Planning Factors

Housing prices are intricately linked with broader socioeconomic dynamics and long-term urban planning policies. Governments and municipalities influence land use, infrastructure development, and zoning, all of which alter real estate value over time.

Zoning laws, for instance, determine how land may be used—residential, commercial, mixed-use, or green space. Properties near upcoming commercial zones often appreciate in value faster due to anticipated increases in accessibility and employment opportunities. However, predictive models may fail to account for zoning changes if datasets are outdated.

Another important factor is infrastructural development. Projects like new highways, metro lines, or even bike paths can significantly increase property desirability. These investments improve accessibility and reduce commuting times, which are strong predictors of price appreciation. Including planned infrastructure projects in predictive models, though difficult, could enhance their foresight.

Socioeconomic indicators such as literacy rates, employment trends, average income levels, and crime statistics also shape demand. Urban planners may prioritize redevelopment projects in economically disadvantaged areas, spurring gentrification—a process where lower-income residents are displaced as housing becomes more expensive.

A forward-looking model would ideally integrate economic planning data, regional development strategies, and census updates. Predictive accuracy, in this case, moves beyond property features and embraces broader contextual awareness.

3. Real-Time Pricing vs Historical Modeling

Most predictive models are retrospective—they analyze historical data to forecast future values. However, in highly dynamic markets, real-time factors such as live demand, traffic patterns, and ongoing bidding behavior can influence house prices more than past averages.

Real-time modeling introduces the concept of “live variables,” which could include:

Current interest rates from central banks.

Recent price adjustments on neighboring listings.

Foot traffic data around the property (collected via mobile GPS).

Traffic congestion and daily commute times from map APIs.

While integrating real-time data presents technical and privacy challenges, it offers significant value. For instance, if a property is suddenly relisted at a lower price or if many users are viewing it online, this demand signal could be used to dynamically adjust its predicted value.

Moreover, predictive models could shift from one-time forecasting to continuous valuation. A live model might update its estimate every week as new listings, economic indicators, and buyer behavior change. This evolution leads to predictive systems that are proactive, not just reactive.

The shift from historical to real-time models requires new infrastructure: data streaming, cloud computation, and continuous retraining. While challenging, this advancement would transform predictive modeling from an offline process to an online service.

4. Post-Prediction Value Engineering: Recommendation Systems

Predicting a house price is valuable, but offering actionable advice post-prediction can further empower users. One such approach is “value engineering” through recommendation systems that suggest targeted improvements to enhance a property’s market value.

For example, suppose a model predicts a property's current value at ₹85 lakh. A recommendation engine could analyze comparable properties and suggest upgrades—such as kitchen renovation or solar panel installation—that may increase the home's resale value by 10–15%.

This concept turns a passive estimate into a prescriptive strategy. It bridges data science and real estate consulting. Homeowners, investors, and realtors can use such systems to make data-backed decisions about renovations, staging, or timing of sale.

Key components of such systems include:

ROI estimation for different types of upgrades.

Cost vs value analysis using regional contractor data.

Peer property comparison to identify gaps.

From a machine learning standpoint, this system would require a separate model trained on before-and-after pricing of renovated homes. It must also factor in regional tastes—what adds value in Mumbai may not apply in Jaipur.

This value-engineering layer transforms your ML model into a holistic decision-support platform, improving its commercial and practical appeal.

5. Modeling Gentrification and Urban Displacement

Gentrification is a socio-economic process where urban neighborhoods experience transformation due to an influx of wealthier residents, often resulting in rising property values and displacement of existing communities. Predictive modeling can play a crucial role in understanding and forecasting this complex urban phenomenon.

Traditional models might predict that an area with historically low prices will continue to perform poorly. However, early signs of gentrification—such as rising rental demand, new business openings (cafés, coworking spaces), or increased building permits—may indicate upcoming price appreciation. Failure to capture these signals can lead to underestimation of future values.

A theoretical approach to modeling gentrification would include:

Monitoring demographic shifts (e.g., median age, income).

Tracking commercial development permits and business turnover.

Assessing changes in public sentiment using social media or news coverage.

Studying shifts in political attention and investment.

By creating an index of “gentrification risk” or “urban transition potential,” a model could forecast not just prices, but social dynamics. Such forecasts could inform urban planners, non-profits, and government bodies aiming to preserve affordability or provide relocation assistance.

On the flip side, this raises ethical dilemmas. Predictive models could be used to preemptively buy up property in areas forecasted to gentrify, contributing to the displacement the model detects. This highlights the responsibility in designing socially conscious algorithm

Next Steps

Let me know if you'd like me to continue with detailed theory for the remaining five advanced topics:

- Cognitive Biases in Human Valuation vs AI Models
- The Explainability Paradox in High-Stakes ML
- Causal Inference vs Correlation in Housing Data
- GeoAI: The Future of AI + Geospatial Intelligence
- Global vs Local Models: A Scalability Dilemma

PROJECT LIMITATIONS

1. **Data Avail*ability*:** The accuracy and effectiveness of the prediction model heavily rely on the availability and quality of historical sales data. Limitations may arise if there are gap so rin consistencies in the dataset ,which could impact the model' sability to make accurate predictions.

2. **Scope of Drug Categories:** The prediction model's accuracy may vary depending on the diversity and granularity of drug categories included in the dataset .If certain drug categories are underrepresented or absent, the model's performance for those categories may be compromised.

3. **Seasonal Variations**: house sales often exhibit seasonal patterns influenced by factors like weather, holidays, or disease out breaks.While the model may account for seasonality to some extent, predicting sales accurately during periods of unusual or unpredictable variations can be challenging.

4. **Market Dynamics:** The real-estate market is dynamic, with new drugsentering the market, patents expiring, and changes in treatment guidelines. These market dynamics can impact sales trends and require continuous monitoring and adjustment of the prediction model to remain relevant.

5. **Dependency on User Input Quality:** The accuracy of predictions is contingent on the quality and accuracy of user-provided inputs such as start date ,end date ,and drug category. Incorrect or ambiguous inputs may lead to erroneous predictions.

6. **Data Privacy and Compliance:** House sales data often contain sensitive information about patients, healthcare providers, and proprietary business practices. Ensuring compliance with data privacy regulations (e.g., GDPR, HIPAA) and protecting confidential information may impose constraints on data collection, sharing, and analysis.

7. **Regulatory Compliance**: The prediction model must adhere to regulatory guide lines and compliance standards applicable to house sales forecasting. Failure to comply with regulations may result in legal or ethical consequences.

8. **Uncertain Future Events:**Predicting house sales involves fore casting future events and trends, which inherently carry uncertainty. Unforeseen events such as regulatory changes, breakthroughs in medical research, or unexpected market disruptions can challenge the accuracy of predictions and require ongoing adaptation of the model.

5.Appendices

Index.html file

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>House Price Prediction</title>
    <!-- Add your CSS styles or include external stylesheets here -->
    <style>
        /* Add your styles here */
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f4f4f4;
        }

        header {
            background-color: #333;
            color: #fff;
            padding: 10px;
            text-align: center;
        }

        main {
            max-width: 800px;
            margin: 20px auto;
            padding: 20px;
            background-color: #fff;
            box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
        }

        footer {
```

```
        text-align: center;
        padding: 10px;
        background-color: #333;
        color: #fff;
        position: fixed;
        bottom: 0;
        width: 100%;
    }

    /* Add styling for the form */
    form {
        margin-top: 20px;
    }

    label {
        display: block;
        margin-bottom: 8px;
    }

    select {
        width: 100%;
        padding: 8px;
        margin-bottom: 16px;
    }

    button {
        background-color: #333;
        color: #fff;
        padding: 10px;
        border: none;
        cursor: pointer;
    }

    /* Style for displaying the predicted price */
    #predictedPrice {
        margin-top: 20px;
        font-weight: bold;
    }
</style>
```

```
</head>
<body>
    <header>
        <h1>House Price Prediction</h1>
    </header>
    <main>
        <p>Welcome to House Price Prediction Model!</p>

        <!-- Form for input fields -->

<form id="predictionForm">
    <label for="beds">Bedrooms:</label>
    <select id="beds" name="beds">
        <option value="" disabled selected>Select number of bedrooms</option>
        {% for bedroom in bedrooms %}
            <option value="{{ bedroom }}">{{ bedroom }}</option>
        {% endfor %}
    </select>

    <label for="baths">Baths:</label>
    <select id="baths" name="baths">
        <option value="" disabled selected>Select number of bathrooms</option>
        {% for bathroom in bathrooms %}
            <option value="{{ bathroom }}">{{ bathroom }}</option>
        {% endfor %}
    </select>

    <label for="size">Size:</label>
    <select id="size" name="size">
        <option value="" disabled selected>Select size of the house</option>
        {% for house_size in sizes %}
            <option value="{{ house_size }}">{{ house_size }} sqft</option>
        {% endfor %}
    </select>

    <label for="zip_code">Zip Code:</label>
    <select id="zip_code" name="zip_code">
        <option value="" disabled selected>Select zip code</option>
        {% for zip_code in zip_codes %}
```

```
            <option value="{{ zip_code }}">{{ zip_code }}</option>
        {% endfor %}
    </select>

    <!-- Predict Price button -->
    <button type="button" onclick="sendData()">Predict Price</button>

    <!-- Space for displaying predicted price -->
    <div id="predictedPrice"></div>
</form>

    </main>

    <!-- Add your JavaScript scripts or include external scripts here -->
    <script>
        // JavaScript function to fetch options for dropdowns
        function fetchOptions(endpoint, dropdownId) {
            fetch(endpoint)
                .then(response => response.json())
                .then(data => {
                    const dropdown = document.getElementById(dropdownId);
                    dropdown.innerHTML = '<option value="" disabled selected>Select an
option</option>';
                    data.forEach(option => {
                        const optionElement = document.createElement('option');
                        optionElement.value = option;
                        optionElement.textContent = option;
                        dropdown.appendChild(optionElement);
                    });
                });
        }

        // Fetch options for each dropdown on page load
        window.onload = function() {
            fetchOptions('/bedrooms', 'beds');
            fetchOptions('/bathrooms', 'baths');
            fetchOptions('/sizes', 'size');
            fetchOptions('/zip_codes', 'zip_code');
        };
```

```
        // JavaScript function to send data and receive predicted price
        function sendData() {
            const form = document.getElementById('predictionForm');
            const formData = new FormData(form);

            fetch('/predict', {
                method: 'POST',
                body: formData
            })
            .then(response => response.text())
            .then(price => {
                document.getElementById("predictedPrice").innerHTML = "Price: INR " +
price;
            });
        }
    </script>
</body>
</html>
```

Main.py python file

Flask is used here

```
from flask import Flask, render_template, request, jsonify
import pandas as pd
import pickle

app = Flask(__name__)
data = pd.read_csv('final_dataset.csv')
pipe = pickle.load(open("RidgeModel.pkl", 'rb'))

@app.route('/')
def index():
    bedrooms = sorted(data['beds'].unique())
    bathrooms = sorted(data['baths'].unique())
    sizes = sorted(data['size'].unique())
    zip_codes = sorted(data['zip_code'].unique())
```

```
    return render_template('index.html', bedrooms=bedrooms, bathrooms=bathrooms,
sizes=sizes, zip_codes=zip_codes)

@app.route('/predict', methods=['POST'])
def predict():
    bedrooms = request.form.get('beds')
    bathrooms = request.form.get('baths')
    size = request.form.get('size')
    zipcode = request.form.get('zip_code')

    # Create a DataFrame with the input data
    input_data = pd.DataFrame([[int(bedrooms), bathrooms, size, zipcode]],
                              columns=['beds', 'baths', 'size', 'zip_code'])

    print("Input Data:")
    print(input_data)

    # Handle unknown categories in the input data
    # for column in input_data.columns:
    #     unknown_categories = set(input_data[column]) - set(data[column].unique())
    #     if unknown_categories:
    #         # Handle unknown categories (e.g., replace with a default value)
    #         input_data[column] = input_data[column].replace(unknown_categories,
data[column].mode()[0])

    print("Processed Input Data:")
    print(input_data)

    # Predict the price
    prediction = pipe.predict(input_data)[0]
    print(prediction,"*"*20)

    return str(prediction)

if __name__ == "__main__":
    app.run(debug=True, port=5000)
```

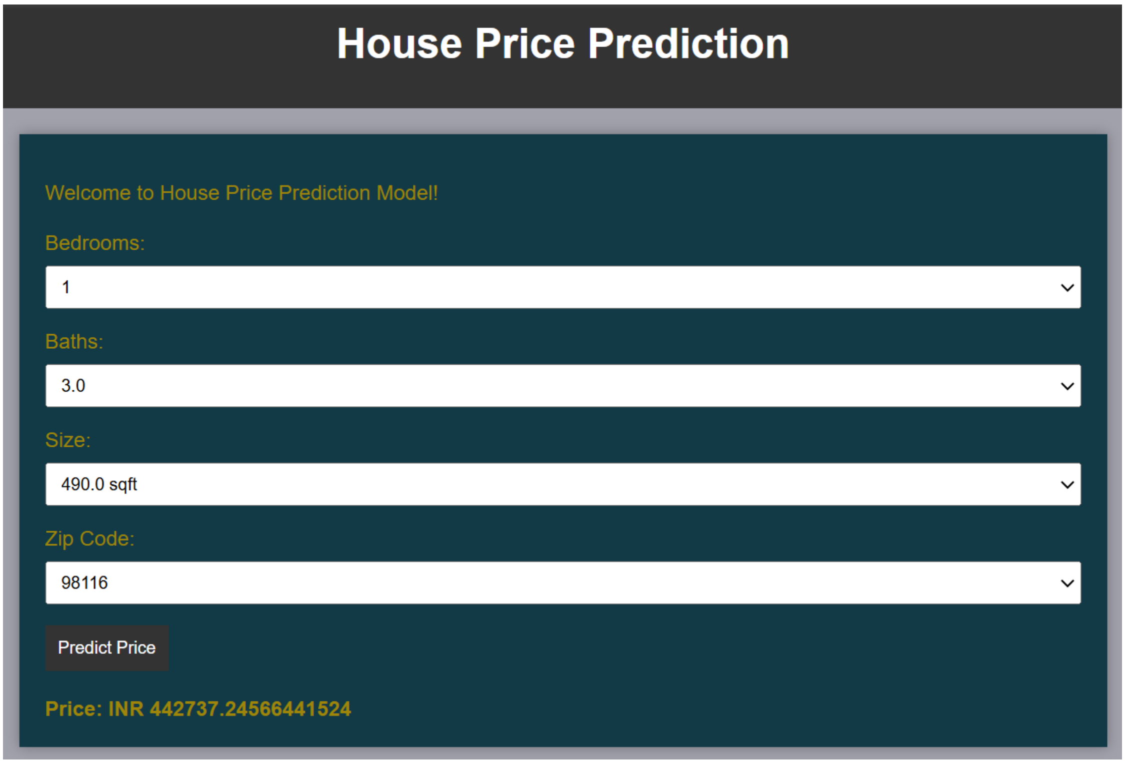


Figure3.29WebPage Display: Prediction Output

Final Web Interface

The above interface represents the final output of the House Price Prediction system, deployed as a user-friendly web application. The design is clean and intuitive, enabling users to easily input property-specific details and instantly receive a predicted house price. The main components of the interface are as follows:

1. Bedrooms:

This dropdown allows the user to select the number of bedrooms in the house. It is one of the key features influencing the property value, as more bedrooms typically indicate a larger, more valuable home.

2. Baths:

This field captures the total number of bathrooms. Like bedrooms, this is a significant feature in determining price, as homes with more bathrooms are generally more desirable and spacious.

3. Size:

This field takes the size of the house in square feet (sqft). Square footage is a direct indicator of the property's area, which is one of the strongest predictors of house price. The input is expected in float format to allow decimal precision.

4. Zip Code:

The location or zip code of the property is a critical determinant of value. Different areas have varying demand, infrastructure, safety levels, and school zones—all of which impact the final price. The user selects the zip code from the dropdown to capture location-based variability.

5. Predict Price Button:

Once all the fields are filled, clicking the “Predict Price” button triggers the machine learning model in the backend. The model processes the input data, makes a prediction, and outputs the estimated house price.

6. Predicted Price Output:

The predicted price is displayed just below the button. In this case, the result is shown as “Price: INR 413869.2468…”, representing the estimated cost of the home based on the inputs. The currency format can be adjusted depending on deployment region (e.g., USD, EUR, INR).

This deployment is built using web development frameworks like Flask or Streamlit (depending on your setup) and serves as an end-to-end integration of the trained machine learning model into a usable, real-world application. It bridges the gap between data science and user accessibility, allowing non-technical users to benefit from predictive analytics in real estate decision-making.

6.CONCLUSION AND FUTURE WORK

Conclusion:

In conclusion, the successful development and deployment of the Polynomial Regression represent a significant milestone in house sales prediction. By allowing users to input start and end dates, as well as select specific drug categories, the application provides tailored predictions to meet individual needs within the real-estate industry.

Through the utilisation of machine learning techniques such polynomial regression, the model generates accurate predictions by analysing historical sales data and identifying patterns and trends. This empowers real-estate companies to make informed decisions regarding inventory management, resource allocation, and sales strategies.

Overall, the House Sales Prediction tool represents a significant advancement in house sales forecasting, offering a combination of advanced analytics and user-centric design to support the dynamic needs of the industry. By harnessing the power of machine learning and deploying it through accessible platforms locally, this solution empowers stakeholders to optimise their operations and drive business growth effectively.

Future Work:

1. Despite the results obtained have been satisfactory, it would be interesting to see if, using a broader Database than that was used in the work, the results obtained could have a precision that could be considered even more acceptable. The inclusion of a broader Database (for example, adding more techniques for data visualisation, analysis) in the sales , the house distribution companies may achieve a greater room for inventory management as well.

2. Explore time series analysis techniques to capture seasonal patterns, trends, and other temporal dependencies in the sales data. Implement advanced time series forecasting models such as ARIMA, SARIMA, Prophet, or LSTM networks to improve the accuracy of sales predictions.

3. Deploy the model on cloud platforms like AWS, Azure ,or Google Cloud Plat form to make it accessible from anywhere

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4. Analytics Vidhya: https://www.analyticsvidhya.com/
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